MIL-D-52645A(ME)
16 May 1973
SUPERSEDING
MIL-D-52645(ME)
19 May 1969

MILITARY SPECIFICATION

DISTILLATION EQUIPMENT SET,

WATER, THERMOCOMPRESSION TYPE:

150 GALLONS PER HOUR,

PACKAGING OF

1. SCOPE

1.1 This specification covers the requirements for materials, methods, and procedures for the preservation, packaging, packing, and marking of the 150 gallons per hour (gph) water distillation equipment set (SC4620-97-CL-E04).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.

FSC 4620

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Military

MIL-P-116	-Preservation, Methods of.
MIL-P-514	-Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	-Treatment and Painting of Materiel.
MIL-D-3209	-Distillation Units, Water, Thermo-compression-Type, Packaging of.
MIL-P-10603	-Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-C-11029	-Citric Acid, Technical.
MIL-C-12044	-Chests, Plywood.
MIL-T-14398	-Tanks, Fabric, Collapsible: Water, Nylon, Rubber-Coated.
MIL-L-21260	-Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-D-52386	-Distallation Units, Water, Thermo-compression: Trailer-Mounted, 150 Gallons Per Hour.

STANDARDS

Military

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-Marking for Shipment and Storage.
MIL-STD-731	-Quality of Wood Members for Containers and Pallets.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2) to prove, prior to starting production packaging and packing, that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification. Examination and tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The preproduction

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pack may be accomplished utilizing either the preproduction model equipment set or a production equipment set. If the preproduction model set is utilized, any preservation, packaging, and packing shall be removed by the supplier at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model set and production sets.

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Distillation unit. The distillation unit shall be preserved and packaged in accordance with MIL-D-3209, level A.

3.3.1.2 Adapter, nozzle, pail, scoop, spout, and strainers. The adapter, nozzle, pail, scoop, spout, and strainers shall require no preservation and packaging. When furnished wrapped or packaged, they may remain as received, provided space is available for packing in accordance with the applicable figure herein.

3.3.1.3 Calcium hypochlorite. The bottle of calcium hypochlorite shall be packaged in a polyethylene (inner) bag, and heat sealed. Each sealed "inner bag" shall be overpackaged in a polyethylene (outer) bag and heat sealed. The polyethylene inner and outer bags shall be fabricated from material conforming to L-P-378, type I, class, grade and finish optional, 0.006 inch thickness, in the form of extruded lay-flat tubing without side seams. The end seams shall be heat-sealed not less than 3/8-inch wide with a smooth polished finish. No type of parting agent shall be used in the fabrication of the seams. Sealing for both inner and outer bags shall result in seams having no blisters or voids in the seam, nor shall the bag be punctured or torn.

3.3.1.4 Citric acid. The fifty pound drums of citric acid packaged as specified in MIL-C-11029 shall require no further packaging.

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3.3.1.5 Extension light and gaskets. The entension light shall be coiled or hanked to a minimum dimensions and secured with cotton tape or twine. Gaskets of like description in quantities specified for each set shall be preserved together in accordance with MIL-P-116, method IC-1, or IC-3.

3.3.1.6 Hoses and hose assemblies. The cotton hoses shall be rolled to a minimum size roll and secured with cotton tape or twine. Each hose assembly shall be coiled and the threaded couplings connected. The 1-1/2-inch hoses shall contain two coils. The 1-inch hoses shall contain three coils.

3.3.1.7 Maintenance tools. Maintenance tools such as but not limited to screw drivers, hammer, knife, pliers, grease gun, and wrenches shall be preserved in accordance with PPP-P-40. Packaging is not required.

3.3.1.8 Protector, sound, aural. Each protector shall be packaged in a polyethylene bag as specified in 3.3.1.3.

3.3.1.9 Tanks. The tanks shall be preserved and packaged in accordance with MIL-T-14398.

3.3.1.10 Gasoline cans. Each gasoline can shall be flushed or fogged with preservative lubricating oil conforming to MIL-L-21260, type I, grade 30. Excess oil shall be drained. Packaging is not required.

3.3.1.11 Pumps. The pumps shall be preserved and packaged in accordance with MIL-P-10603, level A.

3.3.2 Level B.

3.3.2.1 Set components excluding the distillation unit. Preservation and packaging shall be as specified for level A.

3.3.2.2 Distillation unit. The distillation unit shall be preserved and packaged in accordance with MIL-D-3209, level B.

3.3.3 Level C.

3.3.3.1 Set components excluding the distillation unit and tanks. Preservation and packaging of the distillation set components shall be limited to the protection deemed necessary by the supplier to provide protection against deterioration and damage from the supplier to the initial destination.

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3.3.3.2 Distillation unit. The distillation unit shall be preserved and packaged in accordance with MIL-D-3209, level C.

3.3.3.3 Tanks. The tanks shall be preserved and packaged in accordance with MIL-T-14398.

3.4 Packing. Packing shall be level A, B, or C as specified (see 6.2).

3.4.1 Level A.

3.4.1.1 Set components excluding the distillation unit, citric acid, pumps, and tanks. Packing of the set components shall be in the chest specified in figures 1 through 3, with the components located in accordance with figure 4.

3.4.1.2 Distillation unit. The distillation unit shall be packed in accordance with MIL-D-3209, level A, and as specified in MIL-D-52386.

3.4.1.3 Pumps. The pumps shall be packed in accordance with MIL-P-10603, level A. The pumps shall be shipped separate from, but at the same time as, the distillation set components.

3.4.1.4 Tanks. The tanks shall be packed in accordance with MIL-T-14398.

3.4.1.5 Citric acid. The two drums of citric acid shall be packed side by side in a snug fitting box conforming to PPP-B-601, overseas type, style I, grade optional. The box shall be strapped with two girthwise straps conforming to QQ-S-781, type I or IV, class B, 3/8-inch X 0.020.

3.4.2 Level B.

3.4.2.1 Set components excluding the distillation unit, citric acid, pumps, and tanks. Packing of the distillation set components shall be as specified in 3.4.1.1.

3.4.2.2 Distillation unit. The distillation unit shall be packed in accordance with MIL-D-3209, level B. The cap as specified in MIL-D-52386 for level A shall not be required.

3.4.2.3 Pumps. The pumps shall be packed in accordance with MIL-P-10603, level B. The pumps shall be shipped separate from, but at the same time as, the distillation set components.

3.4.2.4 Tanks. The tanks shall be packed in accordance with MIL-T-14398.

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3.4.2.5 Citric acid. The citric acid shall be packed as specified in 3.4.1.5 except that the box shall be domestic type, and strapping shall be class A.

3.4.3 Level C.

3.4.3.1 Set components excluding the distillation unit, citric acid, and tanks. The distillation set components shall be packed as specified in 3.4.1.1.

3.4.3.2 Distillation unit. The distillation unit shall be packed in accordance with MIL-D-3209, level C.

3.4.3.3 Tanks. The tanks shall be packed in accordance with MIL-T-14398.

3.4.3.4 Citric acid. Citric acid packaged as specified in 3.3.1.4 shall require no further packing.

3.5 Unitization.

3.5.1 Level A. The boxed distillation set components (except pump), and boxed fabric tanks shall be unitized in accordance with figure 5.

3.5.2 Level B. Unitization shall be as specified for level A except that strapping shall be class A.

3.6 Marking. Marking shall be in accordance with MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a completed pack of any one level prepared for shipment shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the defects marked "X" for the applicable level in table I. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. Packs shall be examined for the defects listed in table I.

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TABLE I. Classification of Defects

Defect	Level A	Level B	Level C
101. Materials, methods, or containers not as specified. Each incorrect material, method or container shall constitute one defect.	X	X	-
102. Distillation unit not preserved, packaged, and packed as specified.	X	X	X
103. Calcium hypochlorite bottle not overpacked as specified.	X	X	-
104. Citric acid not packaged as specified.	X	X	-
105. Extension lights and gaskets not packaged as specified.	X	X	-
106. Cottom hoses not rolled to minimum size roll and secured as specified.	X	X	X
107. Rubber hose assemblies not coupled as specified.	X	X	-
108. Maintenance tools not preserved as specified.	X	X	-
109. Sound protector not packaged as specified.	X	X	-
110. Tanks not preserved, packaged, and packed as specified.	X	X	-
111. Gasoline cans not preserved as specified.	X	X	-
112. Pumps not preserved in accordance as specified.	X	X	X
113. Marking, illegible, incorrect, incomplete, or missing.			

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or orders for the preservation, packaging, packing, and marking in section 5 of MIL-D-52386 for type I or type II 150 gph water distillation equipment sets. For army procurement, reference to the equipment set is that described in Army Supply Catalog SC4620-97-CL-E04.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

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(b) Time frame required for submission of the preproduction pack (3.1).
(c) Level of preservation and packaging and level of packing required (see 3.3 and 3.4). Level B preservation and packaging is intended to provide limited but economical protection and should be specified only when it is determined the water distillation sets will be held in covered storage.

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the water distillation equipment sets in accordance with this specification.

Custodian:

Army - ME

User activity:

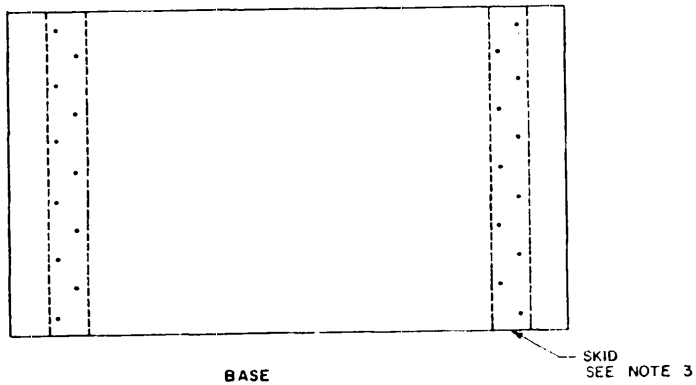
Army - SM

Preparing activity:

Army - ME

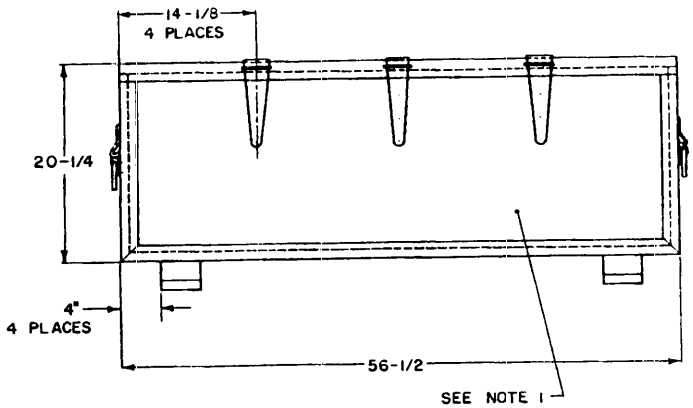
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NOTES:

1. CHEST SHALL BE CONSTRUCTED IN ACCORDANCE WITH MIL-C-12044, USING .75 NOM PLYWOOD IN ACCORDANCE WITH NN-P-530, GROUP A OR B, WITH THE EXCEPTIONS SPECIFIED HEREON AND CORNER COVER PULLS AND LOADING PLAN TO BE OMITTED.
2. TREAT AND PAINT IN ACCORDANCE WITH MIL-T-704, TYPE B.
3. SKIDS SHALL BE FABRICATED OF NOMINAL 1 X 4 AND NOMINAL 2 X 4 LUMBER CONFORMING TO REQUIREMENTS FOR CLASS 3, NON-STRUCTURAL MEMBERS OF MIL-STD-731. SKIDS SHALL BE ATTACHED TO BASE OF CHEST BY MEANS OF #12 FLAT HEAD WOOD SCREWS, 2-3/4" IN LENGTH, IN A PATTERN, AS SHOWN, WITH SCREWS IN TWO ROWS ON 6" CENTERS.
4. ALL DIMENSIONS ARE IN INCHES.



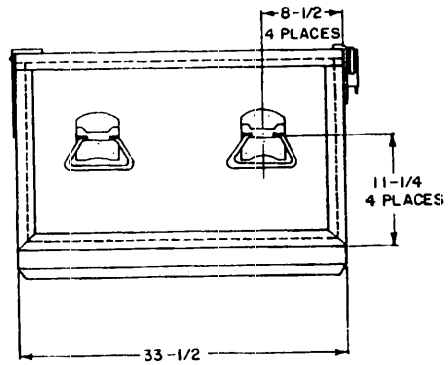


Figure 1. CHEST

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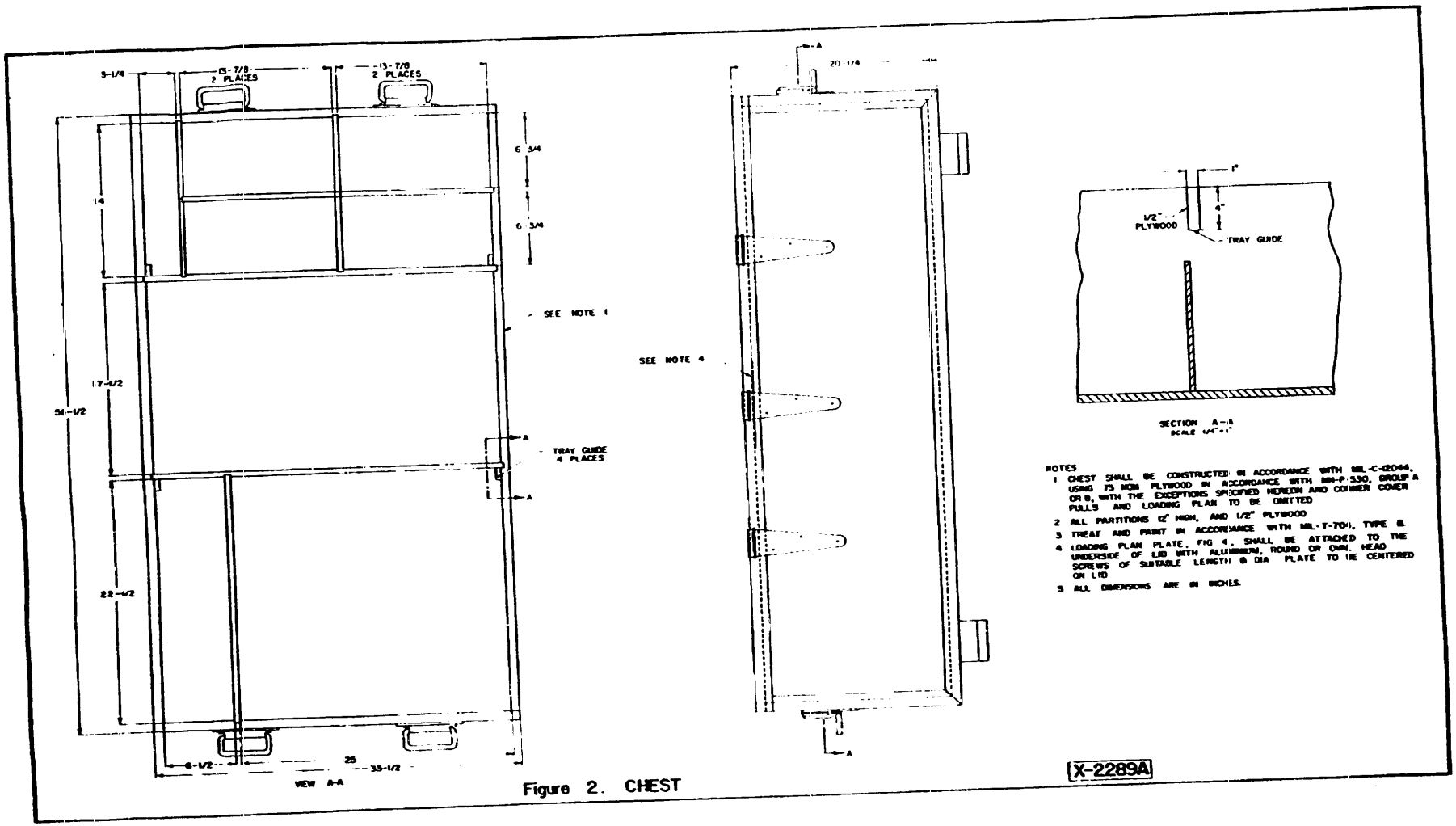


Figure 2. CHEST

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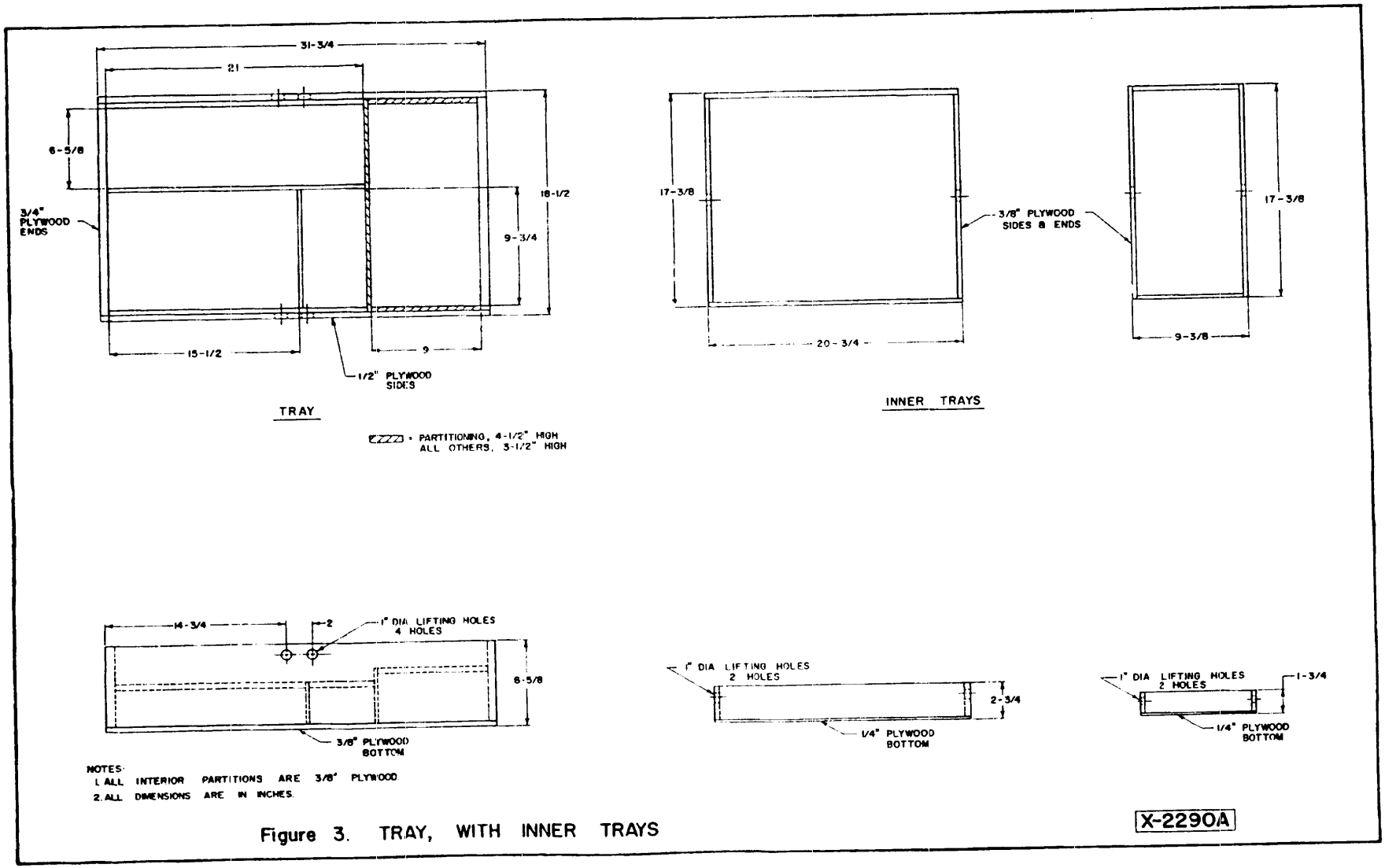
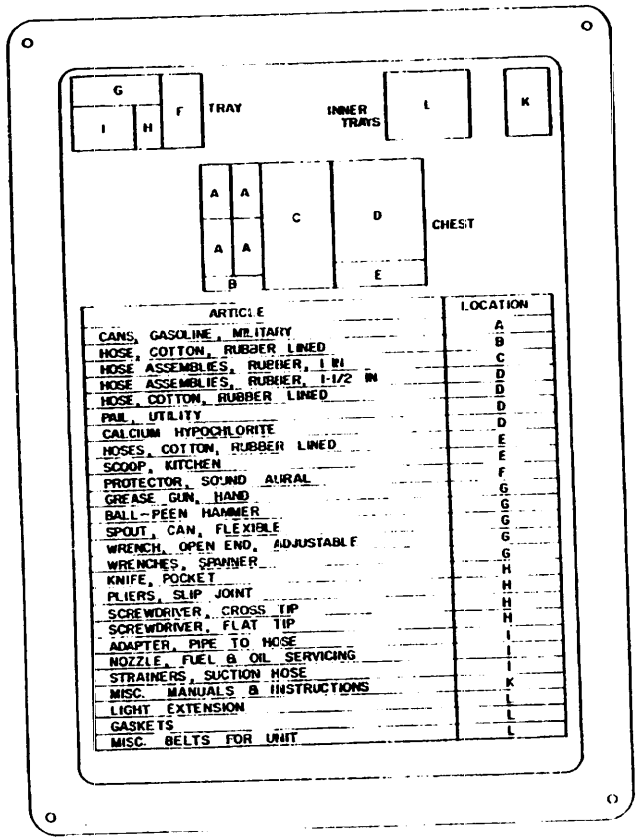


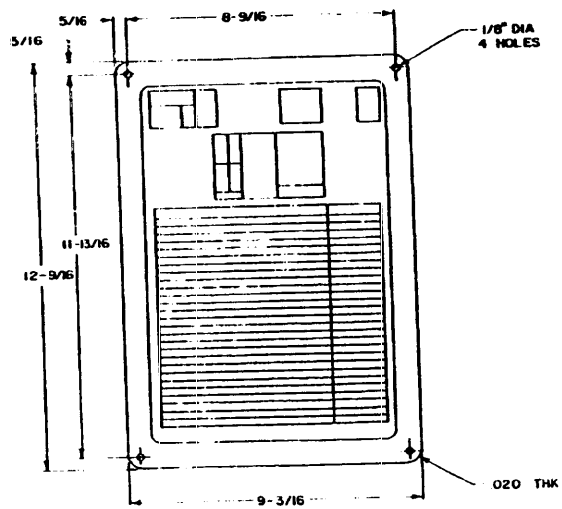
Figure 3. TRAY, WITH INNER TRAYS



ARTICLE	LOCATION
CANS, GASOLINE, MILITARY	A
HOSE, COTTON, RUBBER LINED	B
HOSE ASSEMBLIES, RUBBER, 1 IN	C
HOSE ASSEMBLIES, RUBBER, 1-1/2 IN	D
HOSE, COTTON, RUBBER LINED	D
PAIL, UTILITY	D
CALCIUM HYPOCHLORITE	E
HOSES, COTTON, RUBBER LINED	E
SCOOP, KITCHEN	F
PROTECTOR, SOUND AURAL	G
GREASE GUN, HAND	G
BALL-PEEN HAMMER	G
SPOUT, CAN, FLEXIBLE	G
WRENCH, OPEN END, ADJUSTABLE	G
WRENCHES, SPANNER	H
KNIFE, POCKET	H
PLIERS, SLIP JOINT	H
SCREWDRIVER, CROSS TIP	H
SCREWDRIVER, FLAT TIP	I
ADAPTER, PIPE TO HOSE	I
NOZZLE, FUEL & OIL SERVICING	I
STRAINERS, SUCTION HOSE	K
MISC. MANUALS & INSTRUCTIONS	L
LIGHT EXTENSION	L
GASKETS	L
MISC. BELTS FOR UNIT	L

VIEW FOR PHOTOGRAPHING
SCALE 1/1

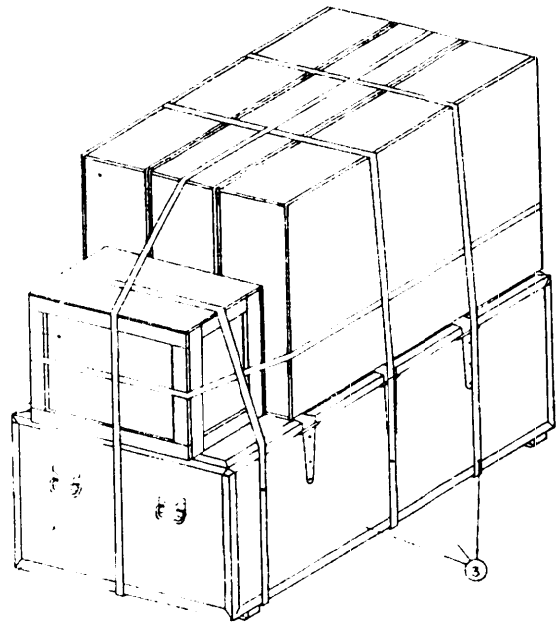
NOTES
1. LOCATE CHARACTERS AND LINE APPROXIMATELY AS SHOWN.
2. PLATE SHALL CONFORM TO MIL-P-514 TYPE III, COMPOSITION C, OF TYPE I, GRADE A, CLASS 1 MATERIAL.
3. ARTICLES LISTED ARE FOR USER'S INFORMATION.
4. ALL DIMENSIONS ARE IN INCHES.



X-229IA

Figure 4. LOADING PLAN PLATES

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NOTE
STRAPPING PATTERNS FOR INDIVIDUAL CONTAINERS NOT SHOWN

FIND NO.	NOMENCLATURE
1	500 GAL. TANKS, 3 EA., BOXED
2	SET CHEST
3	STRAPPING TO CONFORM TO QQ-S-781, TYPE I OR IV, CLASS B, 3/4" X 0.020"
4	BOXED DRUMS OF CITRIC ACID

Figure 5. UNITIZATION OF COMPONENTS

X-2299A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-D-52645A(ME) Distillation Equipment Set, Water, Thermocompression Type: 150 Gallons per Hour, Packaging Of

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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U. S. ARMY MOBILITY EQUIPMENT
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